ABG-VBG Analysis

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```
# TODO (future work):
# P1: ABG residual imbalance >0.10 - consider redefining target population / stratified standardization / overlap
 ↪  restriction.
# P3: Separation/nonconvergence in adjusted outcome models - consider penalized logistic / level collapsing, and
 ↪  define reporting policy.
# TODO consider adding discontinuity (at 45 mmHg CO2 on ABG and 50 mmHg CO2 on VBG; start with raw binning and LOESS
 ↪  to evaluate if there's a jump, then consider regression discontinuity or similar if so).
```

1 Data Pre-Processing and Unweighted Analyses

This code pulls in the master database (a STATA file) and does some initial cleaning - this will only need to be run once, and then the data can be accessed in the usual way.

Package dependencies are managed via `renv` for reproducibility. Use `renv::restore()` before rendering.

1.0.1 Package Set Up

1.1 Helper functions for model diagnostics

Seed escrow for MI and GBM runs

component	seed
Multiple imputation (mice)	20251206
ABG propensity GBM (non-MI)	42
VBG propensity GBM (non-MI)	42
MI PS (glm RCS) seeds (ABG per imputation)	20251206 + imputation index
MI PS (glm RCS) seeds (VBG per imputation)	30251206 + imputation index

Converts the data from a STATA format to rdata if the rdata file does not exist. If it does already exist, it just loads that.

1.1.1 Configuration for the IPW models

Run metadata (pilot vs full)

run_id	run_mode	pilot_frac	mi_ram_gb	mi_ram_gb_source	mi_batch_threshold
20260424_063653	pilot	0.01	16	auto:Darwin	5000

mi_batch_start	mi_batch_start_source	full_n	sampled_n	subset_n
2	auto:auto:Darwin	833476	8335	5175

Codebook artifact path: Results/codebookr.docx. Generation is skipped in pilot mode to keep notebook validation renders fast.

1.1.2 Outcome Variable Creation

1.2 Baseline tables

1.3 Three-level PCO2 categories (unweighted)

Three groups using low/normal/high CO2 categories

1.4 Observed outcome rates by CO2 bin (unweighted)

1.5 Restricted cubic spline regressions (unweighted)

Spline curves are shown as odds ratios relative to CO2_ref (midpoint of the normal range), holding covariates at the reference profile.

1.5.1 Unweighted, Restricted Cubic Spline Regression - ABG by PaCO2

1.5.2 Unweighted, Restricted Cubic Spline - VBG

2 Inverse Propensity Weighting (GBM)

IPW done using Gradient Boosting Methods (GBM) - a type of decision-tree based machine learning. "***Random forests and GBM are designed to automatically include relevant interactions for variables included in the model.*** As such, using a GBM to estimate the PS model, can reduce model misspecification, since ***the analyst is not required to identify relevant interactions or nonlinearities.***" from this citation: PMID: 39947224. PMCID: PMC11825193.

Current propensity score uses `covars_gbm` (demographics, comorbidities, encounter type, vitals, labs) as defined above; in this block only `encounter_type` is explicitly factored before weighting.

Note: for all these, I suggested new GBM adjustments that accomplish the following:

1. Smaller GBM & balance-based stopping (stop.method = "smd.max") → faster fit, avoids over-fitting, lighter tails (which lead to extreme weights that are problematic).
2. Target balance compares weighted treated cohort to the full sample; aim for $|\text{SMD}| < 0.1$.
3. Weight stabilization (divide by mean) mitigates a few huge weights. We use one-sided truncation at very small propensities (caps large weights only).
4. Uses robust variance estimation (e.g. allows the variances to change by PaCO2) for IP-weighted GLM; works with splines via rcs(). This is a bit nuanced but I think good to change even though it adds complexity
5. Deterministic seed ensures result replication.

2.0.1 ABG IPW weighting and diagnostics

GBM tuning is shared across ABG and VBG via `gbm_params` to keep symmetry; update there if needed.

Inverse Propensity-Weighted Logistic Regressions with CO2 predictor represented as a restricted cubic spline.

These are covariate-adjusted outcome models (outcome ~ spline(CO2) + X), fit separately for ABG and VBG cohorts using survey::svyglm with robust (design-based) SEs. Spline curves are shown as odds ratios relative to CO2_ref (midpoint of the normal range).

2.0.2 ABG IPW spline models

Restricting plots bewtween 0.02 and 0.98

2.0.3 ABG IPW spline models (2–98th percentile)

VBG uses the same GBM tuning as ABG (shared `gbm_params`).

2.0.4 VBG IPW weighting and spline models

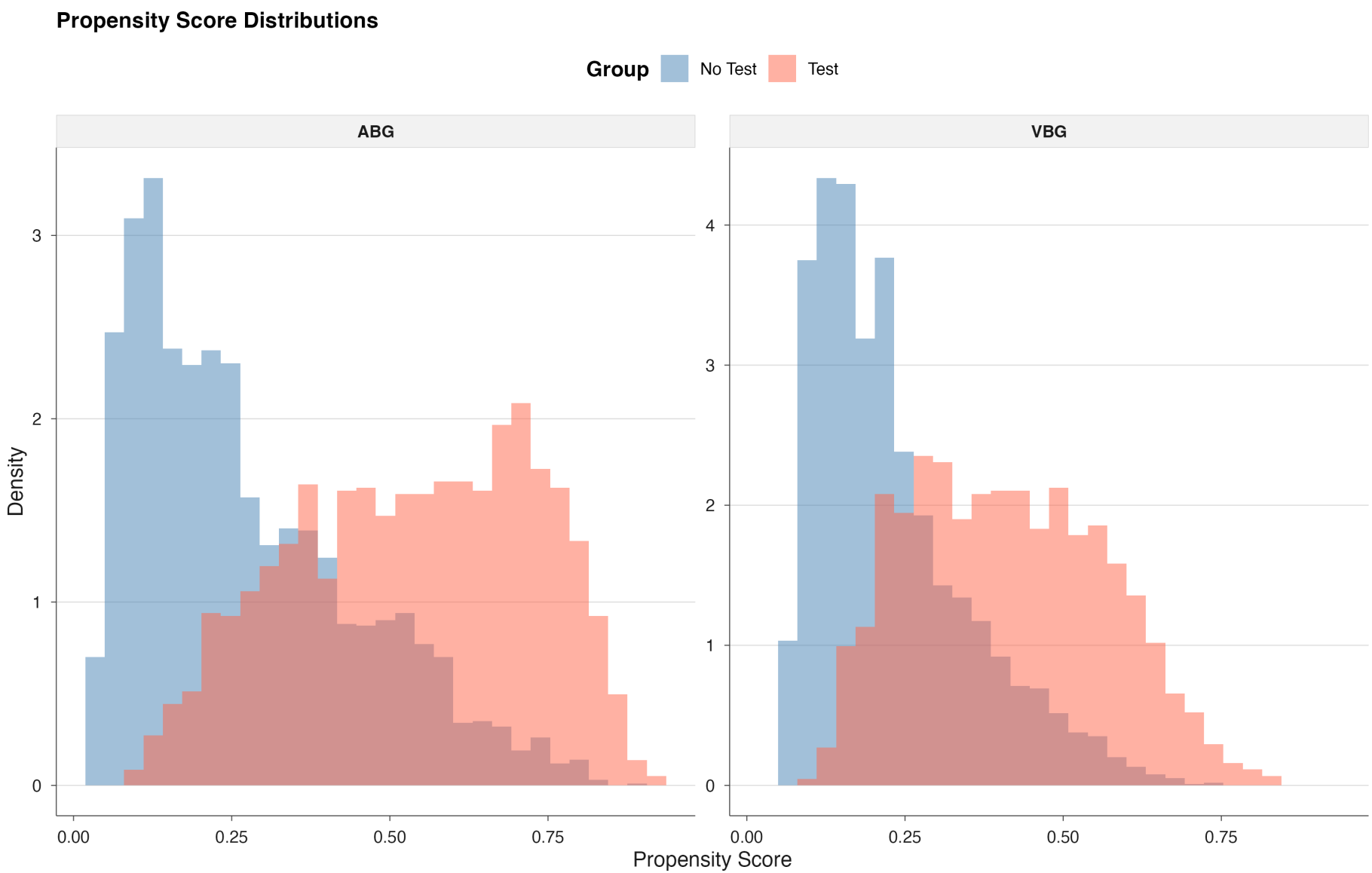
2.0.5 Three-level PCO2 categories (weighted; ABG, VBG)

Three groups with weights and covariate adjustment

2.0.6 Observed outcome rates by CO2 bin (GBM IPSW, non-MI)

2.1 Propensity score diagnostics

Plotting propensity scores


Propensity Score Distributions
Group
No Test
Test
ABG
VBG
Density
0
1
2
3
4
0.00
0.25
0.50
0.75
Propensity Score

3 Multiple Imputation (and logistic IPSW)

added 12/6/2025

3.0.1 Missingness structure and drivers

3.0.2 Monte Carlo error check after MI

3.1 Pre-imputation data prep (consistent types & predictors)

Why: MI models need coherent types; using exactly the same covariates as the propensity score models avoids model drift.

3.2 Imputation model specification (MICE)

3.2.1 Predictor matrix & methods. Run MICE (moderate settings for scale)

3.3 Refit propensity models within each imputation

MI propensity scores use logistic regression with restricted cubic splines (`rms::rcs`, 4 knots by default) for continuous covariates; the same covariate set used in non-MI models is reused here (`covars_ps`). IPSW truncation rules are unchanged.

Note: the MI computations below run in a single pass per imputation (weights, balance, cat3, spline). Subsequent MI sections reuse those outputs and will stop if they are missing.

3.3.1 FAIL-FAST CHECKS

```
            used   (Mb) gc trigger   (Mb) limit (Mb)  max used   (Mb)
Ncells   7084799  378.4   11855017  633.2         NA  11855017  633.2
Vcells 485113151 3701.2  964751313 7360.5      16384 964680566 7360.0
```

```
[1] TRUE
```

3.3.2 ABG propensity (has_abg)

3.3.3 Balance diagnostics across imputations

3.3.4 VBG propensity (has_vbg)

3.3.5 VBG balance

3.4 Weighted outcome models within each imputation + pooling

Within each imputation, fit covariate-adjusted CO2 spline outcome models **only in the measured cohort** (has_abg==1 for PaCO2; has_vbg==1 for VBG CO2), using IPSW weights to address nonrandom testing. Curves are pooled pointwise across imputations (Rubin's rules on the log-OR scale) and displayed as odds ratios relative to CO2_ref at a reference covariate profile.

MI-pooled MI-logistic IPSW associations between CO2 category and outcomes (adjusted)

Cohort	Outcome	Low vs normal OR (95% CI)	High vs normal OR (95% CI)
ABG	IMV	1.41 (1.05, 1.90)	1.26 (0.93, 1.71)
VBG	IMV	1.24 (0.77, 2.00)	1.63 (1.02, 2.60)
ABG	NIV	1.04 (0.63, 1.72)	3.05 (1.99, 4.68)
VBG	NIV	0.76 (0.38, 1.52)	2.69 (1.47, 4.90)
ABG	Death (60d)	1.55 (1.07, 2.25)	1.50 (1.06, 2.14)
VBG	Death (60d)	1.10 (0.68, 1.79)	1.10 (0.70, 1.74)
ABG	Hypercapnic RF	1.26 (0.70, 2.26)	9.21 (5.73, 14.81)
VBG	Hypercapnic RF	0.68 (0.30, 1.54)	6.28 (3.50, 11.25)

Weighted survey GLMs adjusted for baseline covariates; weights = MI-specific GLM (RCS) IPW; m = 20 imputations (seed 20251206); reference = Normal.

3.4.1 Helper: fit + extract log-OR and SE from svyglm

3.4.2 VBG: MI pooled spline models (treated cohort only)

3.5 Explainability

3.6 MI three-level PCO2 helpers and checks

3.7 MI + IPW three-level PCO2 (ABG & VBG)

3.7.1 ABG: MI + IPW, three-level PCO2 outcomes

3.7.2 VBG: MI + IPW, three-level PCO2 outcomes

3.7.3 MI-pooled IPW associations (3-level CO2)

3.7.4 Summary: adjusted CO2-category associations across analysis tracks

3.8 Canonical Categorical Key-Results Figures Across Analysis Tracks

3.9 Observed outcome rates by CO2 bin (MI-logistic IPSW)

3.10 Manuscript outputs summary

```
TRUE
  20
```

3.10.1 Visualization: pooled three-level ORs

3.10.2 Visualization: pooled three-level predicted probabilities

3.10.3 Visualization

3.11 Canonical Key-Results Figures Across Analysis Tracks

The following 4×2 figures are the canonical manuscript-facing spline displays across crude, adjusted, imputed, and weighted analyses.

4 Diagnostics and Exports

4.0.1 MI convergence and mixing

```
png
  2
```

4.0.2 MI stability across m

4.0.3 MI maxit sensitivity (sampled)

4.0.4 Balance diagnostics

4.0.5 Outcome diagnostics

4.0.6 Diagnostics summary and audit

4.0.7 Warning and model-fit diagnostics

4.0.8 Notebook chunk runtime log

5 Analysis of the discordance between predicted probabilities and OR for NIV and IMV

This section is diagnostic and additive. It does not change the primary manuscript estimates; instead it reproduces the NIV/IMV probability-OR discordance and checks whether it is explained by reference profiles, probability scale, residual capture/missingness, sentinel procedure capture, support, setting/site, subgroup, or timing structure.

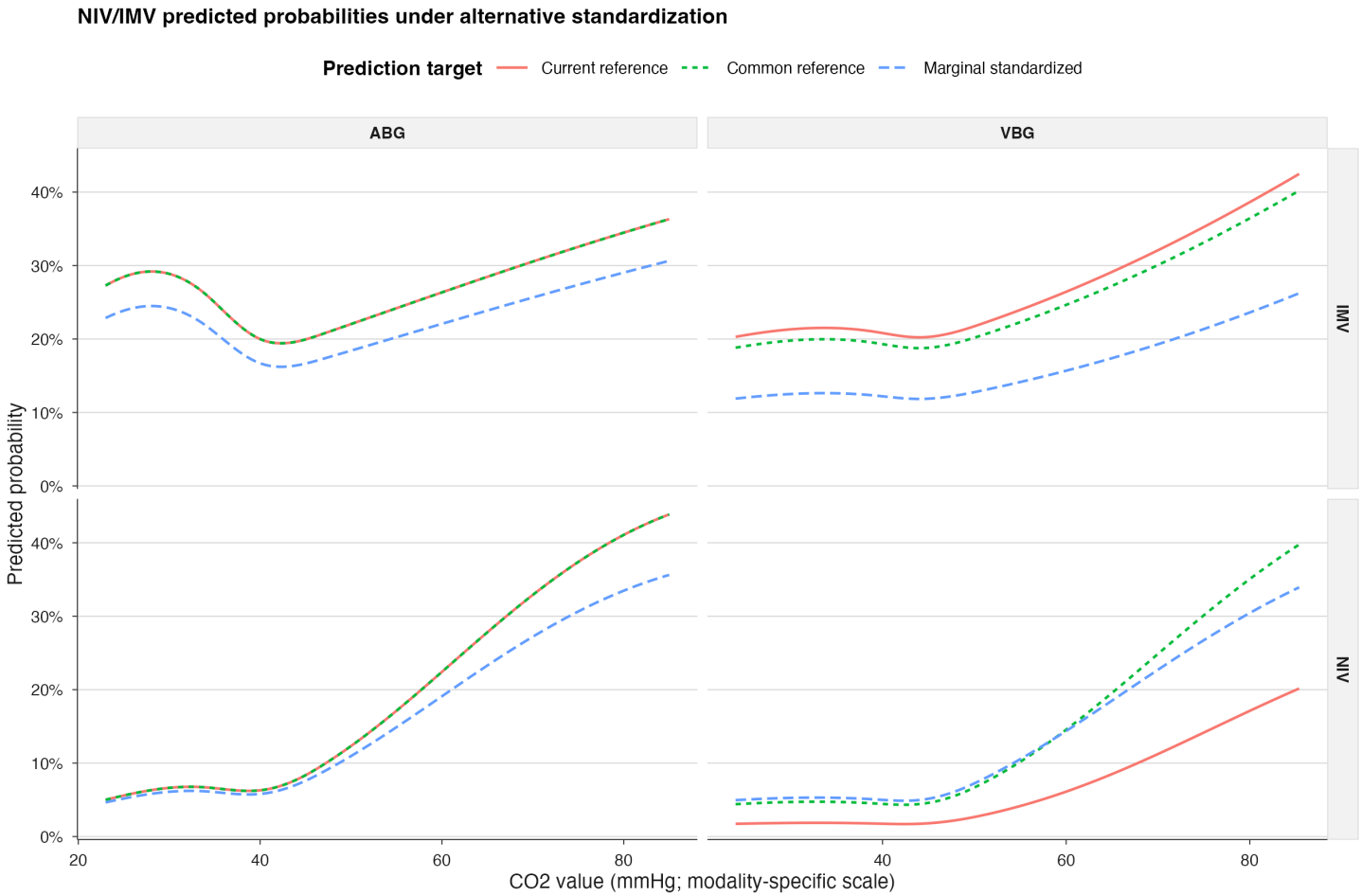
5.1 Pilot validation and scope

Current NIV/IMV OR and predicted-probability summaries

Outcome	Modality	Analysis family	CO2 representation	Low vs normal OR	High vs normal OR
IMV	ABG	MI-logistic IPSW	3-level category	1.413	1.261
IMV	VBG	MI-logistic IPSW	3-level category	1.240	1.631
NIV	ABG	MI-logistic IPSW	3-level category	1.042	3.049
NIV	VBG	MI-logistic IPSW	3-level category	0.760	2.689
IMV	ABG	MI-logistic IPSW	Spline	1.320	1.000
NIV	ABG	MI-logistic IPSW	Spline	1.046	1.371
IMV	VBG	MI-logistic IPSW	Spline	1.033	1.089
NIV	VBG	MI-logistic IPSW	Spline	0.966	1.475

Normal probability	Low probability	High probability	Notes
0.207	0.269	0.248	Current categorical ORs and predicted probabilities.
0.193	0.228	0.280	Current categorical ORs and predicted probabilities.
0.062	0.064	0.167	Current categorical ORs and predicted probabilities.
0.024	0.018	0.062	Current categorical ORs and predicted probabilities.
0.200	0.248	0.200	Current modality-specific reference profile.
0.063	0.066	0.084	Current modality-specific reference profile.
0.203	0.208	0.217	Current modality-specific reference profile.
0.018	0.017	0.026	Current modality-specific reference profile.

[1] TRUE


NIV/IMV predicted probabilities under alternative standardization
Prediction target
Current reference
Common reference
Marginal standardized
ABG
VBG
IMV
NIV
Predicted probability
40%
30%
20%
10%
0%
20
40
60
80
CO2 value (mmHg; modality-specific scale)

[1] TRUE

Tail support and event rates for NIV/IMV

[1] TRUE

Summary of plausible explanations for NIV/IMV probability-OR discordance

Hypothesis	Evidence status	Supported	Not supported	Inconclusive
Reference-profile / standardization effect	Supported	Yes	No	No
Setting or workflow heterogeneity	Supported	Yes	No	No
Differential missingness or procedure capture	Partially supported	No	No	No
Sentinel procedure capture/completeness	Partially supported	No	No	No
Scale / noncollapsibility effect	Not supported	No	Yes	No
Same-encounter timing structure	Not supported	No	Yes	No
Tail support / positivity	Supported	Yes	No	No
Intrinsic modality-specific measurement effect	Not supported	No	Yes	No

Key evidence	Recommended next step
Common or marginal standardization shrank at least one current probability gap by >=25%.	If supported, consider whether manuscript probability curves should use common or marginal standardization in a future methodological ticket.
Setting-rate ranges or IMV-focused setting/site diagnostics show material heterogeneity.	If supported, consider setting-stratified or setting-interaction sensitivity models.
At least one broad capture/completeness proxy differs, but evidence is not specific enough to prove procedure-capture bias.	If supported, audit NIV/IMV source-field construction and coding-density definitions before changing models.
Selected sentinels show some capture/completeness differences, but specificity or directional consistency is insufficient.	If supported, inspect selected sentinel source fields and date-field definitions before attributing NIV/IMV gaps to physiology.
OR-scale agreement was not consistently closer than probability-scale agreement.	Use risk differences/RRs beside ORs when communicating procedure outcomes.
Timing-restricted summaries were feasible but did not show a large difference.	If timestamps become available, restrict to blood gases clearly preceding NIV/IMV.
At least one clinical/quantile bin had ESS <100 or weighted events <10.	If supported, avoid overinterpreting tail probability gaps and consider trimming/support annotations.
Dual-tested subset event rates were generated for direct review.	Compare dual-tested event-rate patterns against primary ABG/VBG gaps before attributing discordance to modality.

Summary of plausible explanations for NIV/IMV probability-OR discordance

- Required discordance diagnostics completed in run mode `pilot` with `pilot_frac = 0.01`.
- Enhanced capture proxies, sentinel v2 specificity notes, IMV setting heterogeneity, timing field mapping, and metadata audits were generated for this diagnostic section.
- Optional timing, first-encounter, dual-tested, and site-specific diagnostics may be marked skipped when required fields or rows are unavailable.
- The final remaining-issues summary separates resolved or strongly clarified findings from unresolved residual IMV, workflow, capture/completeness, and timing questions.
- Detailed artifacts are written under `Results/discordance_*` and `Results/figs/discordance_*`.

5.2 Save, export, and session info

Validation build status

component	status	detail
overall_build	PASS_WITH_WARNINGS	Final validation status derived from artifact registry, glyph audit, duplicate audit, artifact existence, diagnostics, and manuscript-display checks.
artifact_registry	PASS	0 missing required generated artifact rows.
external_manual_assets	PASS	0 declared external/manual rows.
glyph_audit	PASS	0 unsafe manuscript glyph rows.
duplicate_audit	PASS	0 conflicting duplicate/numbering rows.
artifact_existence	PASS	0 required canonical or alias artifacts missing.
manuscript_display	PASS	Manuscript-display status before post-render PDF text scan; wrapper postflight performs the final PDF-level check.
diagnostics_audit	PASS_WITH_WARNINGS	0 high / 2 medium / 0 low diagnostics issues.

Artifact provenance summary

section	kind	status	source_type	pdf_display	n_assets
draft_only	figure	draft_only	suppressed/deprecated	FALSE	1
main	figure	generated	notebook_generated	TRUE	2
main	table	generated	notebook_generated	TRUE	2
supplement	figure	generated	notebook_generated	TRUE	8
supplement	table	generated	notebook_generated	TRUE	5

Figure 1. Cohort assembly and target population for inverse probability-of-sampling weighting

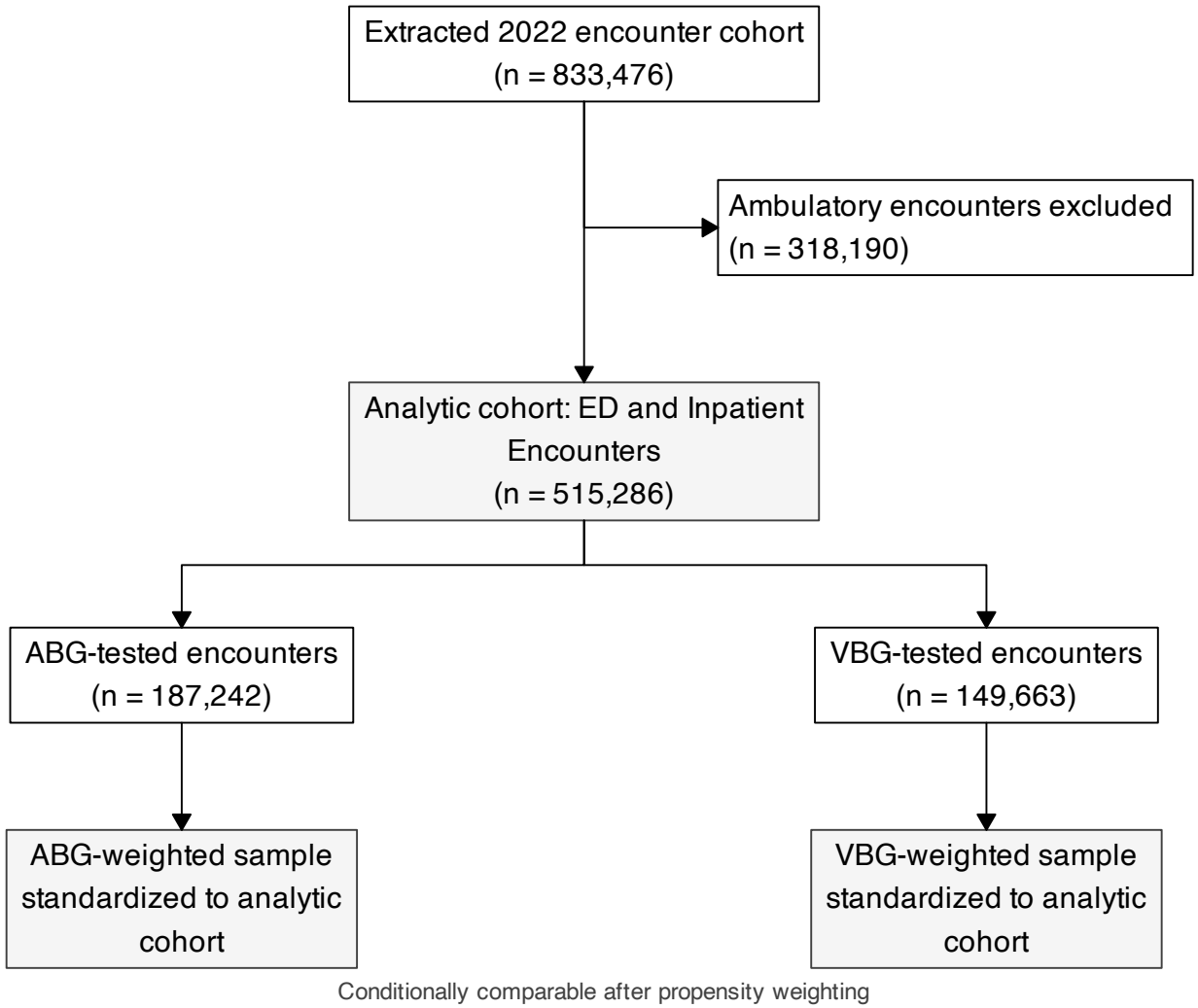
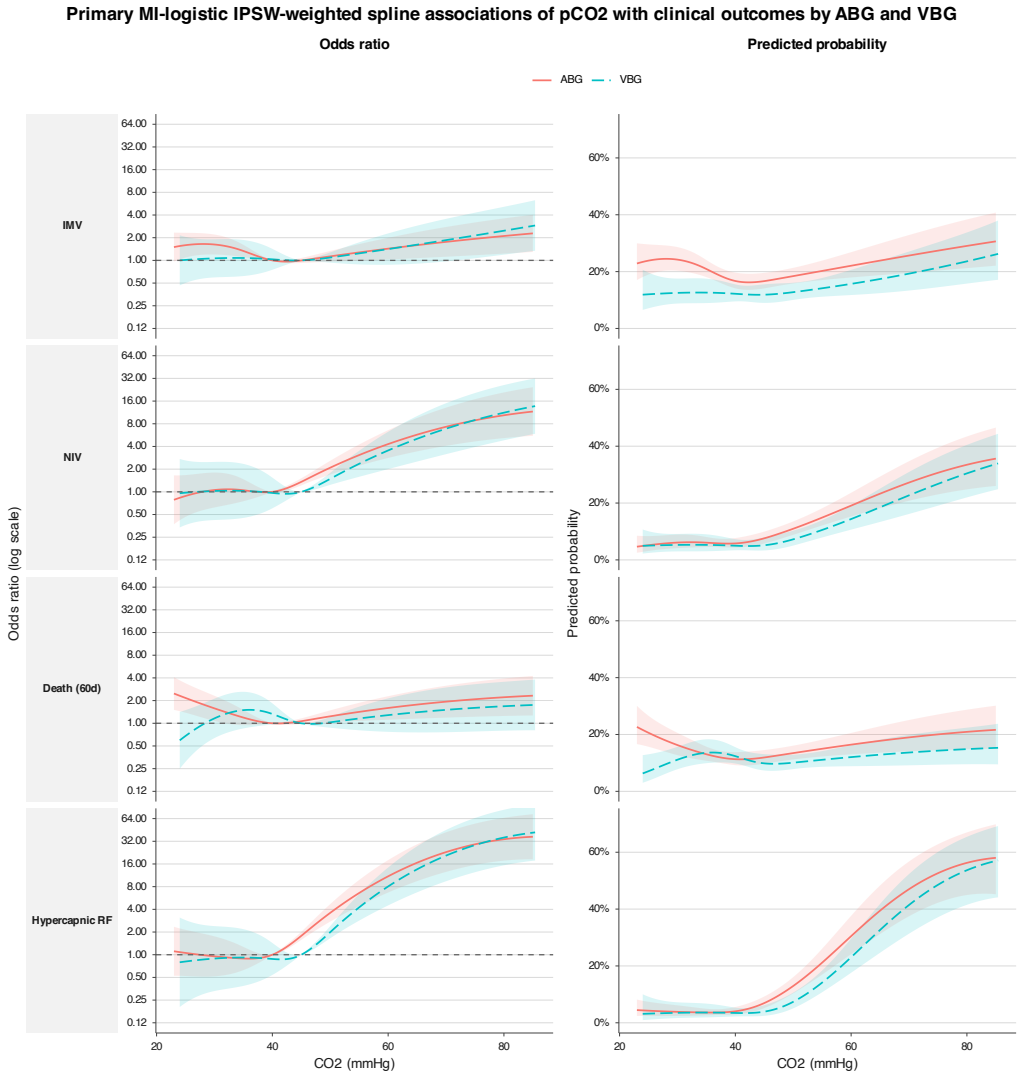


Figure 2. Primary MI-logistic IPSW-weighted spline associations of pCO2 with clinical outcomes by ABG and VBG



Note: These curves come from the primary MI-logistic IPSW analysis. Odds ratio curves are relative to the test-specific reference CO2 values (ABG 40 mmHg; VBG 45 mmHg). Predicted probability curves are marginally standardized to the common eligible source-population covariate distribution.

Table 1. Baseline characteristics of the analytic cohort overall and by receipt of arterial and venous blood gas testing

Characteristic	Everyone	Did not get ABG	Did get ABG	Did not get VBG	Did get VBG
Age	59.5 ± 17.6	58.4 ± 17.7	61.3 ± 17.2	59.6 ± 17.6	59.0 ± 17.7
BMI	31.3 ± 8.5	32.6 ± 8.9	28.8 ± 7.0	31.9 ± 8.6	28.9 ± 7.5
Sex					
Female	2,591 (50%)	1,737 (53%)	854 (45%)	1,921 (52%)	670 (46%)
Male	2,584 (50%)	1,527 (47%)	1,057 (55%)	1,809 (48%)	775 (54%)
Race/Ethnicity					
White	3,280 (63%)	2,009 (62%)	1,271 (67%)	2,513 (67%)	767 (53%)
Black or African American	884 (17%)	601 (18%)	283 (15%)	616 (17%)	268 (19%)
Hispanic	334 (6.5%)	250 (7.7%)	84 (4.4%)	221 (5.9%)	113 (7.8%)
Asian	83 (1.6%)	46 (1.4%)	37 (1.9%)	54 (1.4%)	29 (2.0%)
American Indian	36 (0.7%)	11 (0.3%)	25 (1.3%)	13 (0.3%)	23 (1.6%)
Pacific Islander	10 (0.2%)	8 (0.2%)	2 (0.1%)	8 (0.2%)	2 (0.1%)
Unknown	548 (11%)	339 (10%)	209 (11%)	305 (8.2%)	243 (17%)
Location					
South	2,431 (47%)	1,374 (42%)	1,057 (55%)	1,987 (53%)	444 (31%)
Northeast	1,300 (25%)	921 (28%)	379 (20%)	703 (19%)	597 (41%)
Midwest	406 (7.8%)	246 (7.5%)	160 (8.4%)	264 (7.1%)	142 (9.8%)
West	1,038 (20%)	723 (22%)	315 (16%)	776 (21%)	262 (18%)
OSA	879 (17%)	576 (18%)	303 (16%)	648 (17%)	231 (16%)
Asthma	684 (13%)	464 (14%)	220 (12%)	513 (14%)	171 (12%)
COPD	1,013 (20%)	579 (18%)	434 (23%)	731 (20%)	282 (20%)
CHF	1,044 (20%)	587 (18%)	457 (24%)	741 (20%)	303 (21%)
NMD	233 (4.5%)	130 (4.0%)	103 (5.4%)	175 (4.7%)	58 (4.0%)
PHTN	434 (8.4%)	239 (7.3%)	195 (10%)	295 (7.9%)	139 (9.6%)
CKD	904 (17%)	519 (16%)	385 (20%)	628 (17%)	276 (19%)
Diabetes	1,513 (29%)	954 (29%)	559 (29%)	1,034 (28%)	479 (33%)
Encounter Type					
Emergency	1,710 (33%)	1,427 (44%)	283 (15%)	1,247 (33%)	463 (32%)
Inpatient	3,465 (67%)	1,837 (56%)	1,628 (85%)	2,483 (67%)	982 (68%)

Table 2. MI-pooled, MI-logistic IPSW-weighted 3-level categorical results for ABG and VBG

Panel A. Odds ratios

Gas type	Outcome	Analysis N	Events	Low vs normal OR (95% CI)	High vs normal OR (95% CI)
ABG	IMV	1911	463	1.41 (1.05, 1.90)	1.26 (0.93, 1.71)
VBG	IMV	1445	208	1.24 (0.77, 2.00)	1.63 (1.02, 2.60)
ABG	NIV	1911	183	1.04 (0.63, 1.72)	3.05 (1.99, 4.68)
VBG	NIV	1445	100	0.76 (0.38, 1.52)	2.69 (1.47, 4.90)
ABG	Death (60d)	1911	322	1.55 (1.07, 2.25)	1.50 (1.06, 2.14)
VBG	Death (60d)	1445	200	1.10 (0.68, 1.79)	1.10 (0.70, 1.74)
ABG	Hypercapnic RF	1911	206	1.26 (0.70, 2.26)	9.21 (5.73, 14.81)
VBG	Hypercapnic RF	1445	133	0.68 (0.30, 1.54)	6.28 (3.50, 11.25)

Panel B. Predicted probabilities

Gas type	Outcome	Normal probability (95% CI)	Low probability (95% CI)	High probability (95% CI)
ABG	IMV	0.18 (0.15, 0.21)	0.23 (0.19, 0.27)	0.21 (0.18, 0.25)
VBG	IMV	0.11 (0.09, 0.15)	0.13 (0.10, 0.18)	0.17 (0.13, 0.21)
ABG	NIV	0.06 (0.04, 0.08)	0.06 (0.04, 0.09)	0.15 (0.12, 0.19)
VBG	NIV	0.06 (0.04, 0.09)	0.05 (0.03, 0.08)	0.14 (0.10, 0.19)
ABG	Death (60d)	0.11 (0.09, 0.14)	0.16 (0.13, 0.20)	0.16 (0.13, 0.19)
VBG	Death (60d)	0.11 (0.08, 0.14)	0.12 (0.09, 0.15)	0.12 (0.09, 0.15)
ABG	Hypercapnic RF	0.03 (0.02, 0.05)	0.04 (0.03, 0.06)	0.23 (0.19, 0.28)
VBG	Hypercapnic RF	0.05 (0.03, 0.07)	0.03 (0.02, 0.06)	0.22 (0.17, 0.27)

Table S1. Inclusion criteria into the enriched sample

Phenotype input	*Code system*	*Source code/value or extract field*	*Analysis field(s)*	*Rule*	*Source code status*	*Notes*
Arterial blood gas measurement	LOINC/derived laboratory field	*value_highest_4196*	*value_highest_4196 pco2k has_obg*	Scores.cohort enrichment: ABG exposure definition	available extract concept identifier; upstream LOINC code list not available in analysis extract	Imported ABG-related concept field used in the arterial VOI derivation.
Arterial blood gas measurement	LOINC/derived laboratory field	*value_highest_11556*	*value_highest_11556 obg_gh has_obg*	Scores.cohort enrichment: ABG exposure definition	available extract concept identifier; upstream LOINC code list not available in analysis extract	Imported ABG-related concept field retained with the arterial gas measurement set.
Arterial blood gas measurement	LOINC/derived laboratory field	*value_highest_11558*	*value_highest_11558 obg_gh_bicarbonate has_obg*	Scores.cohort enrichment: ABG exposure definition	available extract concept identifier; upstream LOINC code list not available in analysis extract	Imported ABG-related concept field retained with the arterial gas measurement set.
Arterial blood gas procedure	LOINC/procedure/derived flag	*mean_selected_gas_proc mean_art_gas_proc_first_date*	*mean_selected_gas_proc has_art_gas_proc_first_date has_obg*	Scores.cohort enrichment: ABG exposure definition; timing diagnostics	collapsed derived procedure flag; upstream source code list not available in analysis extract	Procedure-date field is an encounter-day offset when present.
Venous blood gas measurement	LOINC/derived laboratory field	*vbg_coh vbg_gh vbg_hco3 vbg_cobat*	*vbg_coh vbg_gh vbg_hco3 vbg_cobat has_vbg*	Scores.cohort enrichment: VBG exposure definition	collapsed derived measurement fields; upstream LOINC code list not available in analysis extract	The de-identified analysis extract stores VBG measurement variables rather than raw-level LOINC code core.
Venous blood gas procedure	LOINC/procedure/derived flag	*mean_venous_vd_proc mean_venous_vd_proc_first_date*	*mean_venous_vd_proc mean_venous_vd_proc_first_date has_vbg*	Scores.cohort enrichment: VBG exposure definition; timing diagnostics	collapsed derived procedure flag; upstream source code list not available in analysis extract	Procedure-date field is an encounter-day offset when present.
Hypercapnic respiratory failure and related respiratory diagnoses	ICD-10/Charlson/derived diagnosis flags	*hypercap_resp_failure resp_arrd_dx other_resp_failure1 resp_abnormality resp_dig_compl RESPFAIL_ch*	*hypercap_resp_failure resp_arrd_dx other_resp_failure1 resp_abnormality resp_dig_compl RESPFAIL_ch*	Scores.cohort enrichment: hypercapnic respiratory failure outcome definition	collapsed derived diagnosis flags; upstream ICD-10/CM code list not available in analysis extract	These fields identify respiratory-failure phenotypes after upstream code grouping.
Non-invasive ventilation procedure	CPT/PCS/procedure/derived flag	*vent_proc vent_proc_first_date VENTNONINV_ch*	*vent_proc vent_proc_first_date VENTNONINV_ch has_vent has_hypercap_dx has_vent_date working_date test_no_dbs_flag_ion_cbs_flag GRESITY_ch*	Scores.cohort enrichment: NIV outcome definition; timing diagnostics	collapsed derived procedure flag; upstream CPT/PCS/procedure code list not available in analysis extract	The date field is an encounter-day offset and is used only for diagnostic timing sensitivity.
Invasive mechanical ventilation procedure	CPT/PCS/procedure/derived flag	*vent_proc vent_proc_first_date*	*vent_proc vent_proc_first_date VENTINVASIVE_ch*	Scores.cohort enrichment: IMV outcome definition; timing diagnostics	collapsed derived procedure flag; upstream CPT/PCS/procedure code list not available in analysis extract	The date field is an encounter-day offset and is used only for diagnostic timing sensitivity.
Any ventilatory support procedure	CPT/PCS/procedure/derived flag	*vent_proc vent_proc_first_date VENTNONINV_ch*	*vent_proc vent_proc_first_date VENTNONINV_ch*	Scores.cohort enrichment: ventilatory support phenotype definition	collapsed derived procedure flag; upstream CPT/PCS/procedure code list not available in analysis extract	Composite ventilatory-support fields combine NIV and IMV-related upstream procedure groupings.

Figure S1. Covariate balance after MI logistic inverse-probability weighting for ABG and VBG test-ordering models

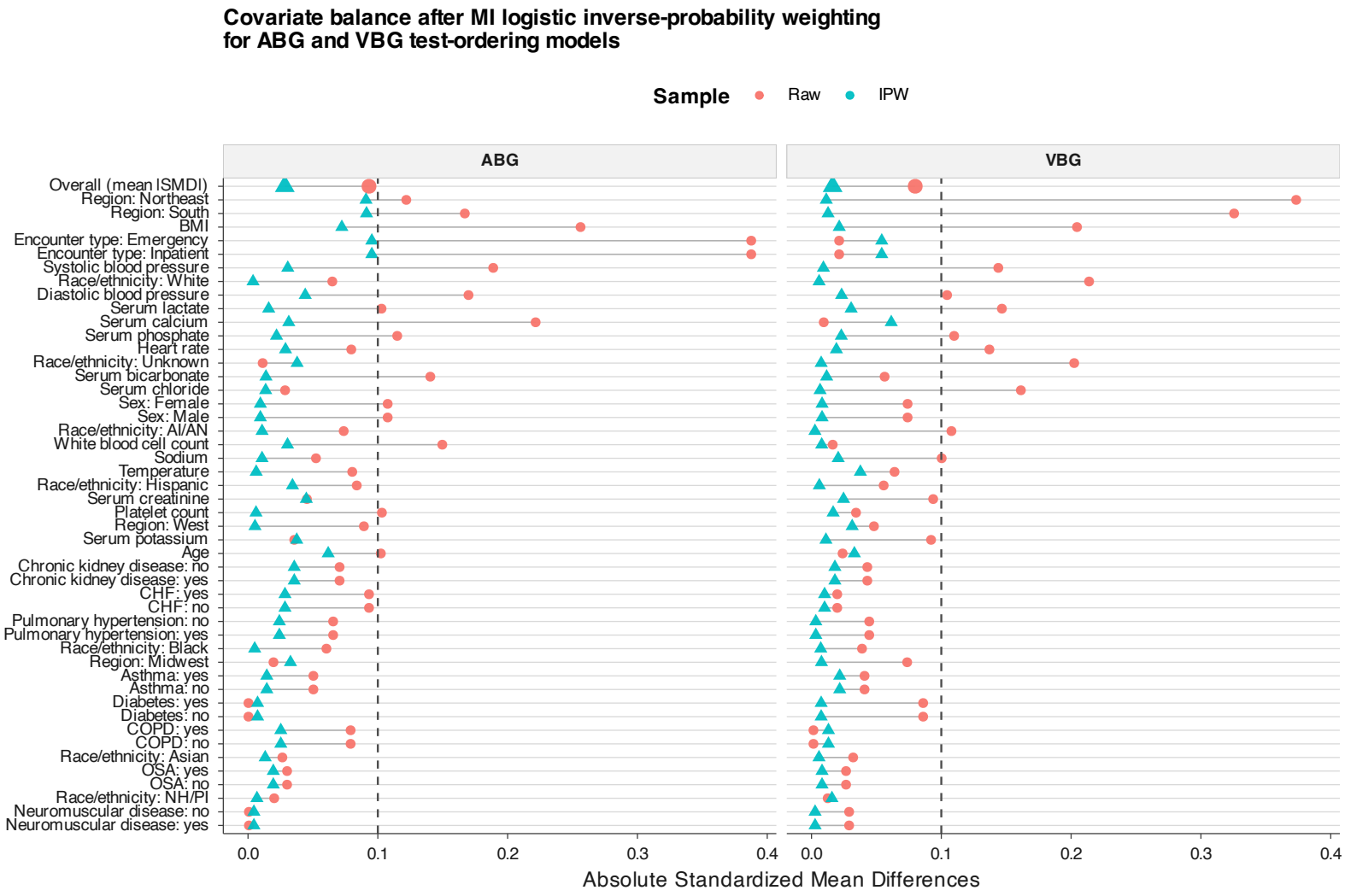


Figure S2. Propensity-score overlap for MI logistic ABG and VBG test-ordering models

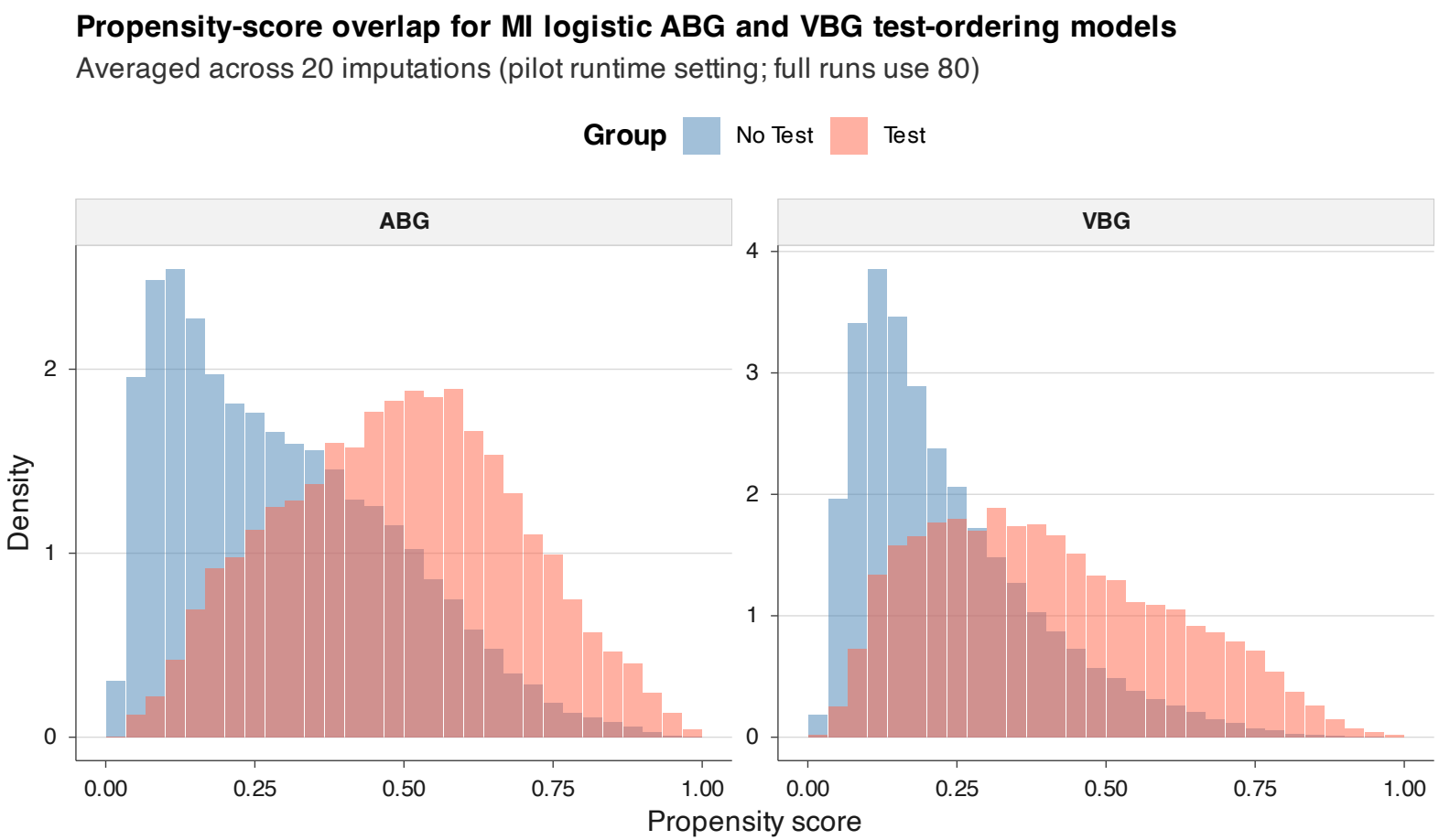


Figure S3. SHAP-style contribution summaries for MI logistic ABG and VBG test-ordering models

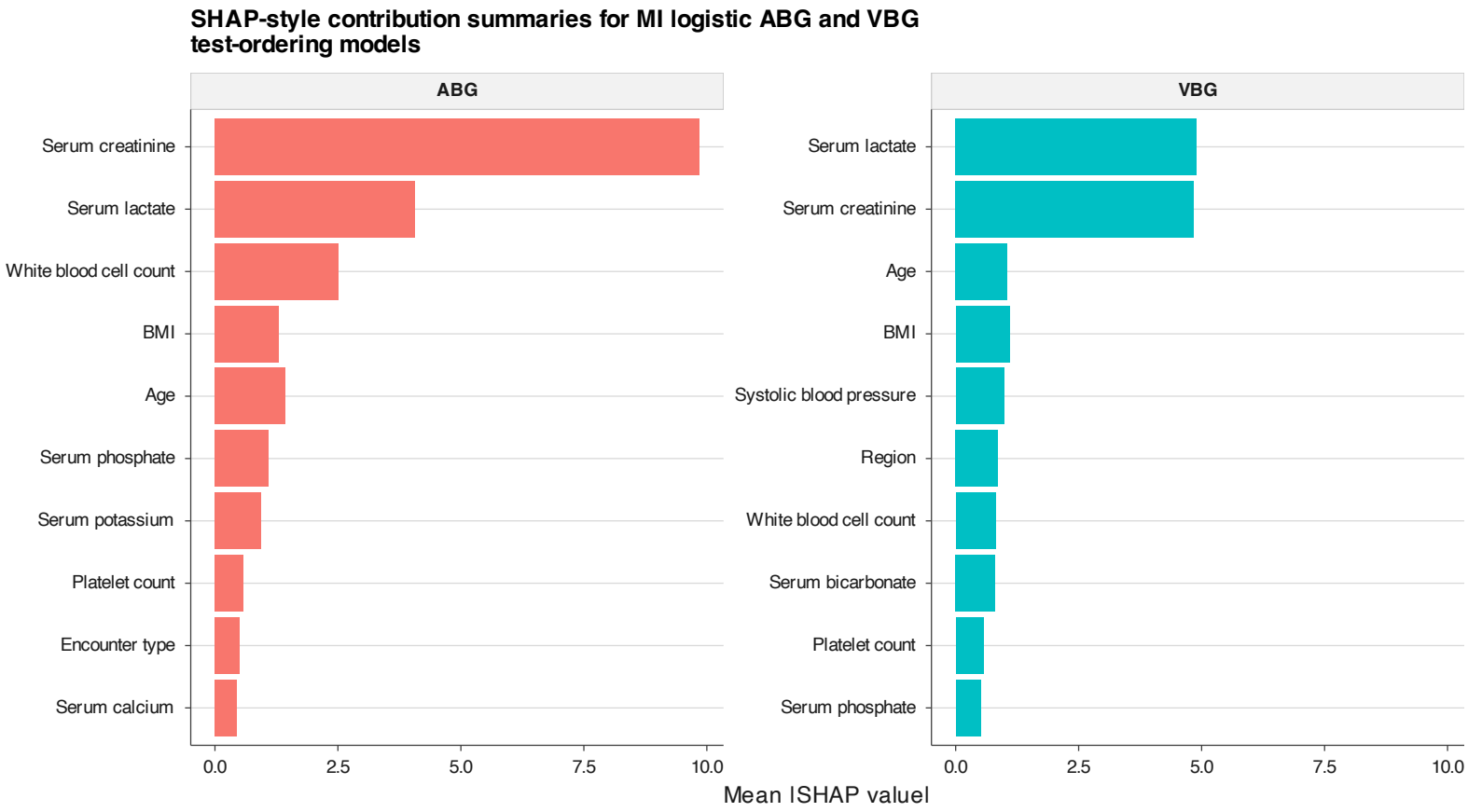
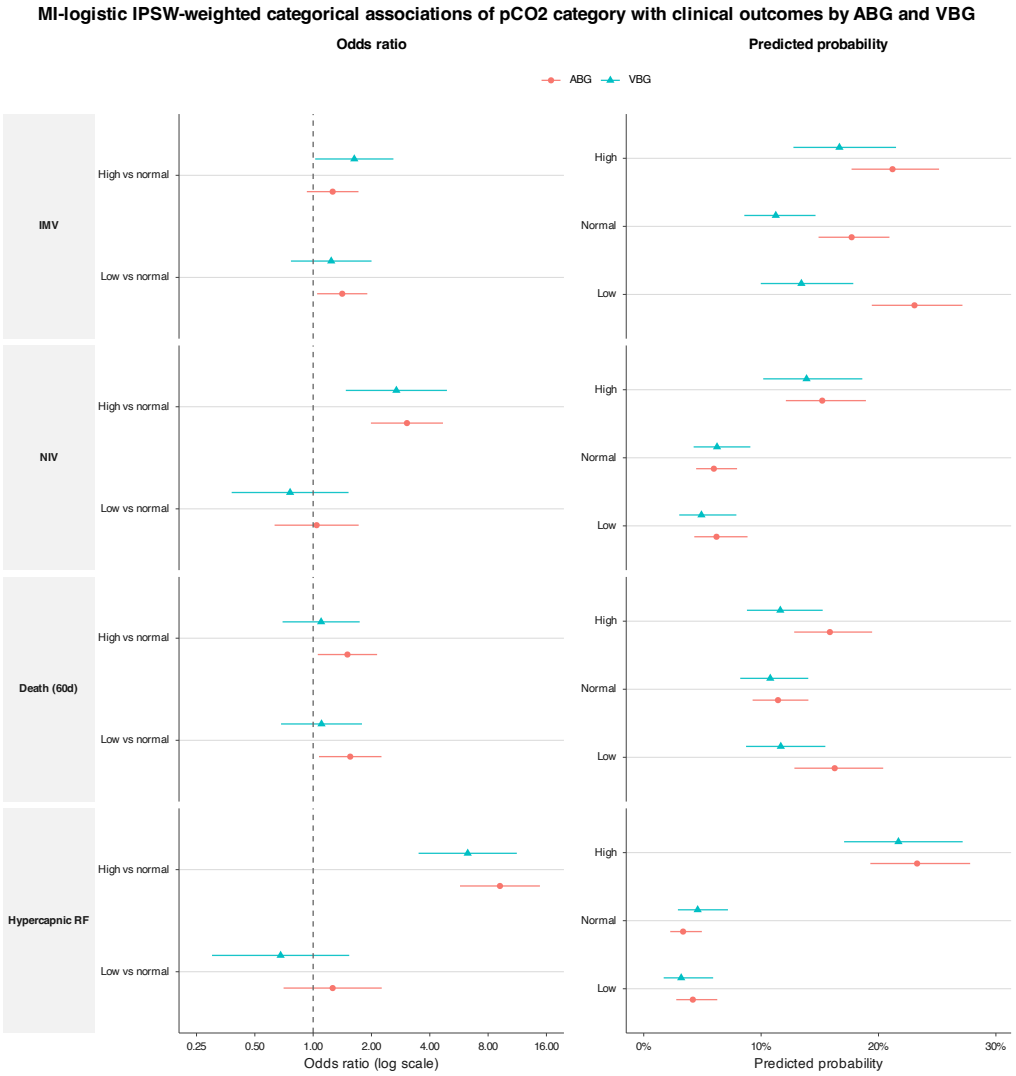
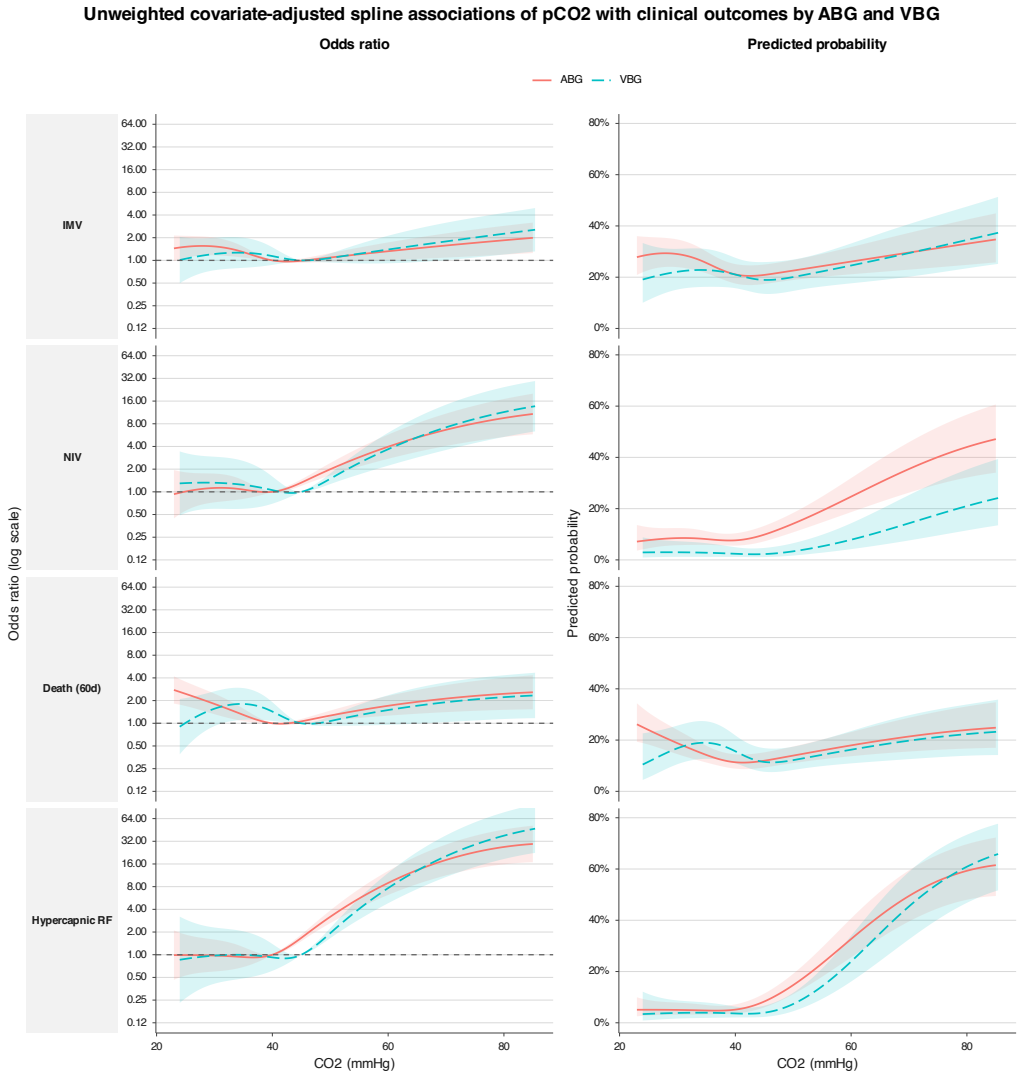


Figure S4. MI-logistic IPSW-weighted categorical associations of pCO2 category with clinical outcomes by ABG and VBG



Note: MI-pooled, MI-logistic IPSW-adjusted categorical odds ratios use the normal CO2 category as the reference. Predicted probabilities are pooled on the link scale and marginally standardized to the common eligible source-population covariate distribution.

Figure S5. Unweighted covariate-adjusted spline associations of pCO2 with clinical outcomes by ABG and VBG



Note: Odds ratio curves are relative to the test-specific reference CO2 values (ABG 40 mmHg; VBG 45 mmHg). Predicted probability curves are conditional on the reference covariate profile X_ref within each test-specific cohort.

Table S2. Crude associations of pCO2 category with clinical outcomes by ABG and VBG

Gas type	Outcome	Contrast	OR (95% CI)
ABG	IMV	Low vs normal	1.44 (1.12, 1.86)
ABG	IMV	High vs normal	1.28 (0.99, 1.65)
VBG	IMV	Low vs normal	1.48 (1.03, 2.13)
VBG	IMV	High vs normal	1.80 (1.27, 2.57)
ABG	NIV	Low vs normal	1.00 (0.65, 1.52)
ABG	NIV	High vs normal	2.26 (1.60, 3.21)
VBG	NIV	Low vs normal	1.21 (0.69, 2.09)
VBG	NIV	High vs normal	2.61 (1.63, 4.23)
ABG	Death (60d)	Low vs normal	1.60 (1.19, 2.15)
ABG	Death (60d)	High vs normal	1.55 (1.16, 2.08)
VBG	Death (60d)	Low vs normal	1.71 (1.19, 2.45)
VBG	Death (60d)	High vs normal	1.58 (1.10, 2.29)
ABG	Hypercapnic RF	Low vs normal	1.36 (0.81, 2.28)
ABG	Hypercapnic RF	High vs normal	8.29 (5.65, 12.50)
VBG	Hypercapnic RF	Low vs normal	0.99 (0.51, 1.85)
VBG	Hypercapnic RF	High vs normal	7.13 (4.58, 11.46)

Table S3. GBM IPSW-weighted associations of pCO2 category with clinical outcomes by ABG and VBG

Gas type	Outcome	Contrast	OR (95% CI)
ABG	IMV	Low vs normal	1.40 (1.06, 1.86)
ABG	IMV	High vs normal	1.14 (0.85, 1.52)
VBG	IMV	Low vs normal	1.34 (0.89, 2.03)
VBG	IMV	High vs normal	1.58 (1.06, 2.37)
ABG	NIV	Low vs normal	1.04 (0.65, 1.69)
ABG	NIV	High vs normal	2.82 (1.90, 4.17)
VBG	NIV	Low vs normal	0.88 (0.47, 1.67)
VBG	NIV	High vs normal	2.77 (1.62, 4.75)
ABG	Death (60d)	Low vs normal	1.52 (1.08, 2.16)
ABG	Death (60d)	High vs normal	1.46 (1.05, 2.02)
VBG	Death (60d)	Low vs normal	1.38 (0.91, 2.10)
VBG	Death (60d)	High vs normal	1.31 (0.87, 1.98)
ABG	Hypercapnic RF	Low vs normal	1.54 (0.84, 2.82)
ABG	Hypercapnic RF	High vs normal	9.39 (5.99, 14.73)
VBG	Hypercapnic RF	Low vs normal	0.76 (0.37, 1.56)
VBG	Hypercapnic RF	High vs normal	7.58 (4.54, 12.67)

Figure S6. Covariate balance after gradient-boosted propensity weighting for ABG and VBG test-ordering models

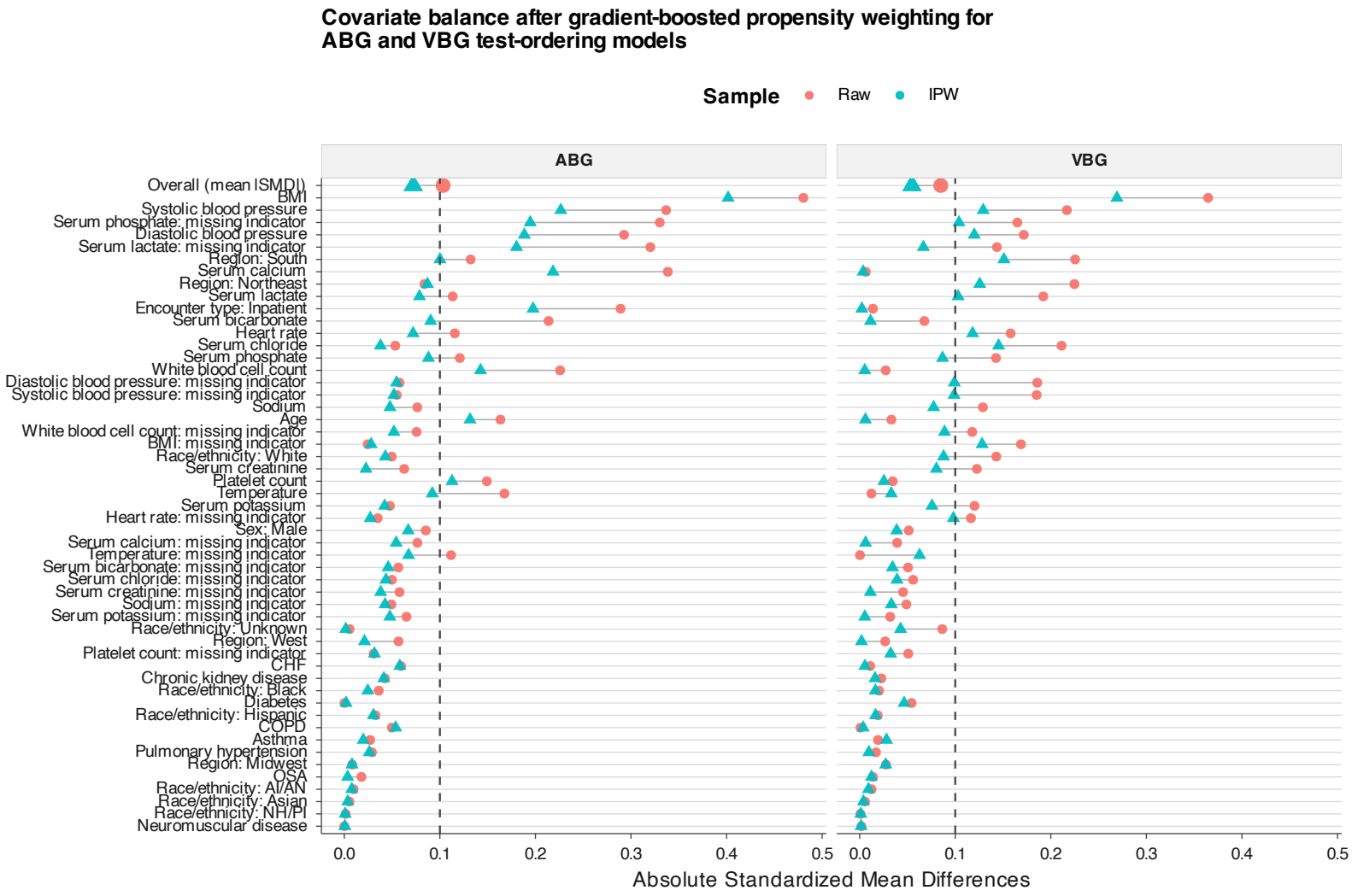


Figure S7. Propensity-score overlap for gradient-boosted ABG and VBG test-ordering models

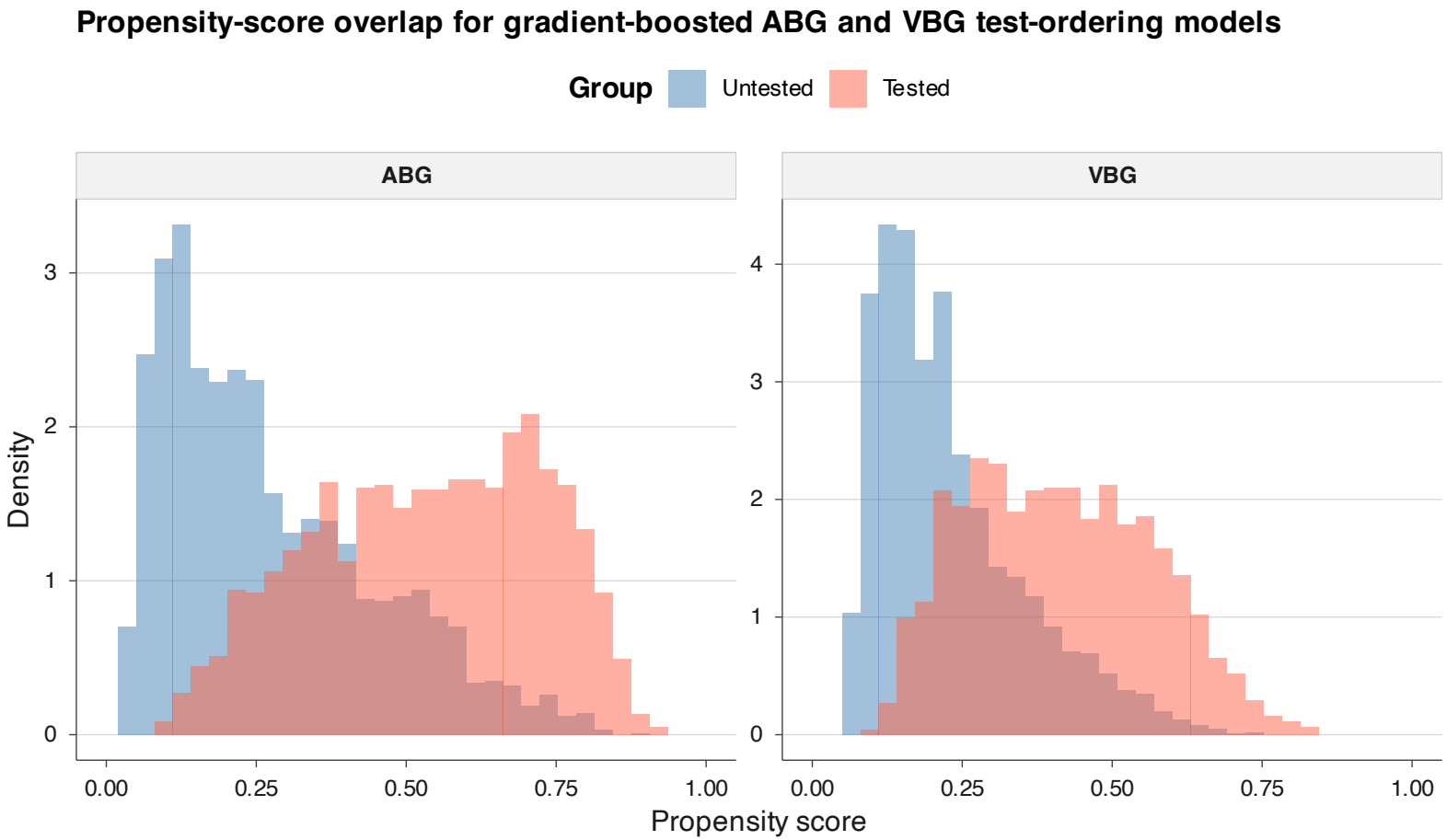


Figure S8. SHAP-style contribution summaries for gradient-boosted ABG and VBG test-ordering models

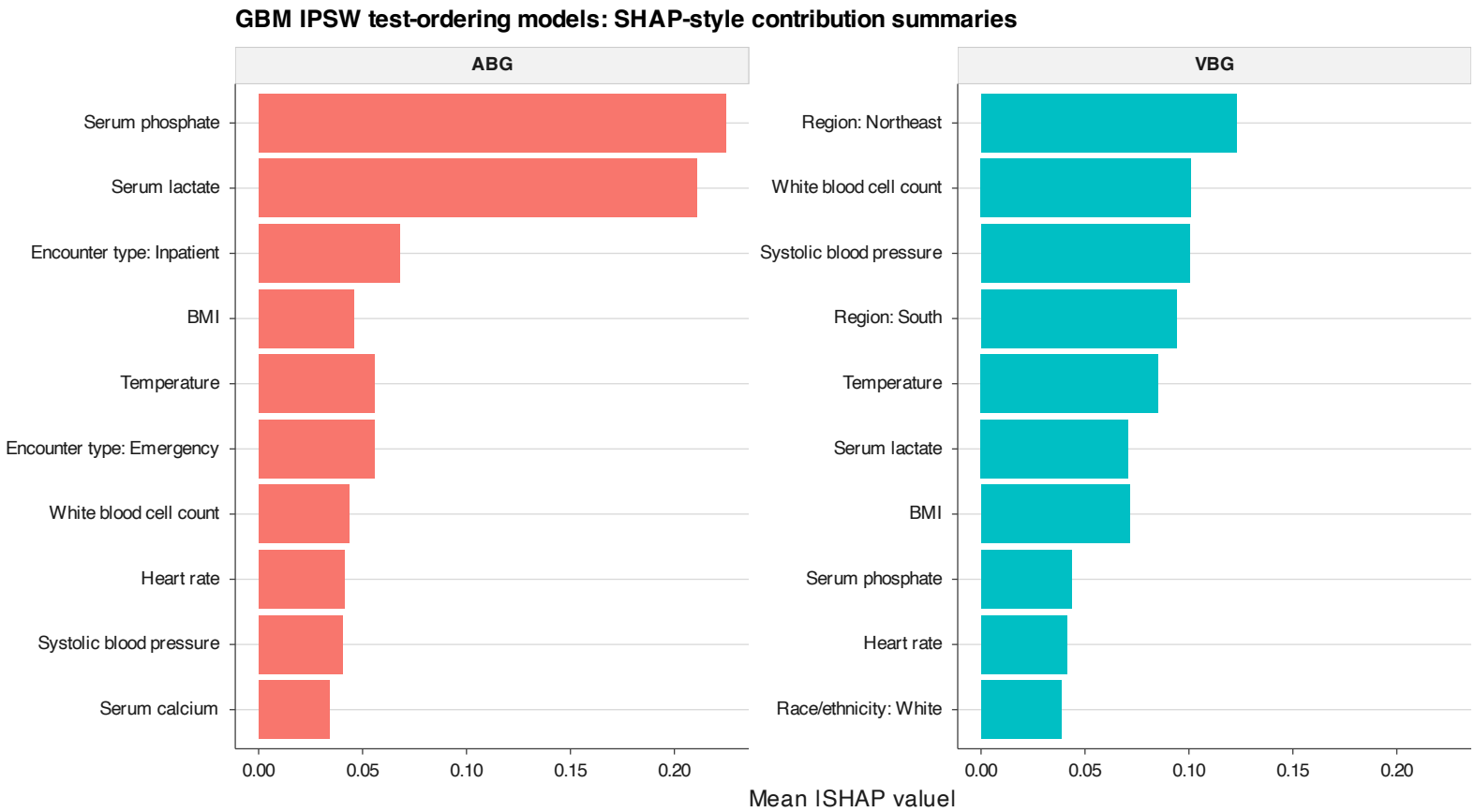


Table S4. Missingness of baseline covariates used in the primary analysis

Variable	Analysis role	% missing	Included in MI model	Imputed if missing	Observed complete	Notes
Age	direct adjustment; PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Sex	direct adjustment; PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Race/ethnicity	direct adjustment; PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
BMI	PS model	56.7%	Yes	Yes	No	Eligible for MICE method: pmm.
COPD	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Asthma	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
OSA	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
CHF	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Neuromuscular disease	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Pulmonary hypertension	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Chronic kidney disease	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Diabetes	PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Location	direct adjustment; PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Encounter type	direct adjustment; PS model	0.0%	Yes	No	Yes	Complete in this render; used as MI predictor or model variable.
Temperature	PS model	48.3%	Yes	Yes	No	Eligible for MICE method: pmm.
Systolic blood pressure	PS model	29.8%	Yes	Yes	No	Eligible for MICE method: pmm.
Diastolic blood pressure	PS model	30.1%	Yes	Yes	No	Eligible for MICE method: pmm.
Heart rate	PS model	36.5%	Yes	Yes	No	Eligible for MICE method: pmm.
Sodium	PS model	5.0%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum creatinine	PS model	9.3%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum bicarbonate	PS model	5.4%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum chloride	PS model	5.8%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum lactate	PS model	60.4%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum potassium	PS model	7.8%	Yes	Yes	No	Eligible for MICE method: pmm.
White blood cell count	PS model	17.3%	Yes	Yes	No	Eligible for MICE method: pmm.
Platelet count	PS model	7.7%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum phosphate	PS model	52.8%	Yes	Yes	No	Eligible for MICE method: pmm.
Serum calcium	PS model	10.0%	Yes	Yes	No	Eligible for MICE method: pmm.

Table S5. Multiple-imputation diagnostic summary

Domain	Metric	Value	Status	Interpretation
Render scope	Current render	pilot (pilot_frac=0.01)	warning	Pilot validation render settings shown (pilot_frac = 0.01); final manuscript Table S5 should be regenerated from the full run.
Render scope	Final manuscript target	full (m=80; maxit=20)	not applicable	Full manuscript settings remain the target for final manuscript-facing diagnostics.
Imputation run	Completed imputations	20	resolved	Imputation count used in the current render.
Imputation run	Iterations per imputation	5	resolved	Configured MICE iteration count for this render.
Imputation run	Variables imputed	16	resolved	Variables with non-empty MICE imputation methods and observed missingness.
Monte Carlo error	Maximum MC error / SE in diagnostic model	0.177	resolved	Diagnostic model ratio; lower values indicate less simulation noise relative to model SE.
Chain diagnostics	Variables with available chain diagnostics	16	resolved	Imputed variables with traceable chain-mean diagnostics.
Chain diagnostics	Variables flagged for drift	14	warning	Variables exceeding drift/slope screening thresholds.
MICE logged events	Total logged events	110	warning	Rows reported by mice::loggedEvents across batches.
Captured warnings	Captured warning rows	0	resolved	Warnings captured by notebook-level diagnostics wrappers.
Outcome fit diagnostics	MI model fits audited	692	resolved	Outcome-model diagnostic rows captured for MI analyses.
Outcome fit diagnostics	Fits with nonconvergence flags	0	resolved	Potential convergence problems requiring review if nonzero.
Outcome fit diagnostics	Fits with explicit separation warnings	0	resolved	Explicit fitted-probability or separation warnings are treated as manuscript-critical failures if present.
Outcome fit diagnostics	Fits with tail-only fitted-probability warnings	540	warning	Tail-only fitted-probability extremes are warnings unless they affect manuscript-facing plotted curves.

Artifact check summary

check_status	severity	allowed_missing	n_rows
present	none	FALSE	68
suppressed_draft_only	none	TRUE	1

Diagnostics audit summary

component	metric	status	severity	value	note
Audit	overall_status	warn	medium	PASS_WITH_WARNINGS	Overall diagnostics audit status derived from the highest-severity unresolved issue.
Balance	abg_max_abs_smd	warn	medium	0.108	ABG target balance threshold is 0.10; pilot/subset threshold misses are validation warnings.
Outcome	separation_flags_total	warn	medium	540	Broad legacy count of separation/extreme-probability flags; tail-only fitted-probability flags are warnings unless manuscript-facing curves are affected.
Outcome	diagnostic_warnings_total	warn	medium	540	Warning count for tail-only or off-profile fitted-probability flags.
Inventory	expected_artifacts_missing	pass	none	0	Count of expected diagnostics artifacts missing from Results/.
MI	smoke_test_failed	pass	none	FALSE	Smoke test failure indicates the MI execution path is unstable.
MI	chain_diagnostics_issue	pass	none	FALSE	numeric_names=FALSE; drift_tail_na_frac=0.000
Balance	vbg_max_abs_smd	pass	none	0.079	VBG target balance threshold is 0.10; pilot/subset threshold misses are validation warnings.
Outcome	diagnostic_failures_total	pass	none	0	Failure count after splitting nonconvergence, explicit separation warnings, and central plotted-curve instability from tail-only probability warnings.
Plotting	plot_drop_log_present	pass	none	TRUE	Plot drop logging should exist even if no rows are present.

Diagnostics audit issues

severity	component	evidence_file	evidence_snippet	why_it_matters	recommended_fix
medium	Balance	Results/balance_target_imp_summary.csv	ABG max|SMD|=0.108	ABG target balance exceeds 0.10 in a pilot/subset render; this is a validation warning, not a full-analysis failure.	Confirm balance on the larger validation subset or full render before changing the weighting model.
medium	Outcome	Results/model_fit_diagnostics.csv	tail/off-profile fitted-probability warnings for 540 / 692 fits	Tail-only fitted-probability extremes require transparent reporting but do not by themselves invalidate manuscript-facing plots.	Keep the warning in the diagnostics summary and reassess if any flagged row affects the displayed central curve range.

run_id	run_ts
20260424_063653	2026-04-24 06:36:53.744157
20260424_063653	2026-04-24 06:36:53.744157

Software: R 4.5.3 ; key packages: mice, WeightIt, cobalt, survey, rms.